Task 6: Deploy Multiple Frontend Apps on Apache with SSL (Subdomain Based Hosting)

Objective

Deploy **any two frontend applications** from the repository below on an Apache web server using **subdomains**, and secure them using **Let's Encrypt SSL certificates**.

Repository:

```
https://github.com/pushkar-iamops/frontend-apps
```

Deployment Requirements

Each application must be hosted using your name as part of the subdomain.

Subdomain Format

- `bookshop-yourname.yourdomain.com`
- `notes-yourname.yourdomain.com`

Example format:

```
appname-yourname.yourdomain.com
```

Each intern will:

- Choose any **two apps**
- Deploy each app on a **separate subdomain**
- Secure both using HTTPS

Prerequisites

- Ubuntu Server (with public IP)
- Domain configured and pointing to server
- DNS A records created for:
  - `appname-yourname.yourdomain.com`
- Ports **80 and 443** open in firewall/security group

Task Instructions

Part 1: Install and Configure Apache

1. Install Apache.
2. Verify Apache is running.
3. Confirm default Apache page loads over HTTP.

Part 2: Clone Repository

1. Clone the repository:

```
https://github.com/pushkar-iamops/frontend-apps
```

2. Choose **any two frontend apps**.
3. Since apps are static, no build step is required.

Part 3: Setup Directory Structure

Create separate directories for each app:

```
/var/www/bookshop-yourname/
/var/www/notes-yourname/
```

Copy respective app files into their directories.

Ensure:

- Proper ownership and permissions are set.
- Apache can read the files.

Part 4: Create Apache Virtual Hosts

Create separate VirtualHost configurations for each subdomain:

Example structure:

```
<VirtualHost *:80>
    ServerName bookshop-yourname.yourdomain.com
    DocumentRoot /var/www/bookshop-yourname
</VirtualHost>

<VirtualHost *:80>
    ServerName notes-yourname.yourdomain.com
    DocumentRoot /var/www/notes-yourname
</VirtualHost>

```

Enable the site configurations and reload Apache.

Verify both apps load over HTTP.

Part 5: Secure with Let's Encrypt SSL

1. Install Certbot with Apache plugin.
2. Generate SSL certificates for both subdomains.
3. Enable automatic HTTP → HTTPS redirection during setup.

After configuration:

- Both apps must load securely over HTTPS.
- Browser must show valid secure padlock.
- No SSL warnings should appear.

Part 6: Verify Auto-Renewal

Run a dry-run renewal test to confirm:

- SSL certificates will renew automatically.

Validation Checklist

Each intern must verify:

- App 1 loads on `https://appname1-yourname.yourdomain.com`
- App 2 loads on `https://appname2-yourname.yourdomain.com`
- HTTP automatically redirects to HTTPS
- Valid SSL certificate installed (no browser warning)
- Auto-renewal test successful

Expected Deliverables

Submit the following:

1. Requirement analysis (Confluence link)
2. Technical documentation (Confluence link)
3. Prepare a short document explaining the risks of not using SSL and how implementing SSL mitigates those risks. (Confluence link)
    - Also include a short video (2–3 mins) explaining the same documentation.

Estimated Completion Time

6 Hours

Reference

How To Secure Apache with Let's Encrypt on Ubuntu | DigitalOcean

HTTPS / SSL via “Let’s Encrypt” on an Apache Web Server